



Pavel Ivashin <ivashin.pasha@gmail.com>

AutoCAD Feature Proposal — Smarter MLEADERALIGN Algorithm with Working AutoLISP Prototype

Писем: 4

Pavel Ivashin <ivashin.pasha@gmail.com>

13 сентября 2026 г. в 00:10

Кому: autocad.beta@autodesk.com

Копия: feedbackcommunity@autodesk.com

Hello AutoCAD Team,

My name is Pavel Ivashin, and I would like to propose an improvement to the standard MLEADERALIGN command in AutoCAD.

I am contacting you by email because access to some Autodesk websites and feedback services is currently unavailable from my region, so I cannot reliably submit the proposal through the normal Autodesk Feedback Community workflow.

While working with assembly drawings containing many MULTILEADER objects, I noticed a limitation of the current MLEADERALIGN command.

The command can align and distribute MLeader content, but it does not appear to take the spatial order of the leader arrowhead points into account when determining the order of annotations.

As a result, the annotations can be distributed correctly while the Leader Lines cross each other, sometimes creating a result that requires significant manual correction.

I originally encountered a similar problem several years ago in CATIA V5 Drafting and developed a VBA algorithm for automatic Balloon alignment.

Recently, I adapted and improved that algorithm for AutoCAD and created a working AutoLISP prototype:

MLeaderSmartAlign

GitHub repository:

<https://github.com/ivashinpavel07/AutoCAD-AutoLISP-MLeaderSmartAlign>

Latest release:

<https://github.com/ivashinpavel07/AutoCAD-AutoLISP-MLeaderSmartAlign/releases/tag/v0.5>

Video demonstration:

<https://youtu.be/xlh3vYKc8fs>

In the video, I first demonstrate the behavior of the standard MLEADERALIGN command and then run MLeaderSmartAlign on the same type of drawing.

The key difference is that MLeaderSmartAlign does not determine the complete annotation order only once.

For every new target position, the algorithm:

1. analyzes all remaining MLeaders;
2. calculates their geometry relative to the current target position and their arrowhead points;
3. selects the most suitable MLeader for that position;
4. places it while preserving its original arrowhead location;
5. removes it from the candidate set;
6. recalculates the geometry of all remaining MLeaders before placing the next one.

For N selected objects, the process is approximately:

$N \rightarrow N-1 \rightarrow N-2 \rightarrow \dots \rightarrow 1$

This iterative recalculation is what significantly reduces crossing Leader Lines.

The current prototype also provides:

- Live Preview while defining the alignment direction;
- preservation of original arrowhead locations;
- equal distribution of MLeader content;
- support for block-based and MText-based MLeaders;
- content-aware alignment;
- cancellation with restoration of the original geometry.

The AutoLISP implementation is intended as a proof of concept. I believe this principle could be implemented more efficiently and robustly directly inside the native MLEADERALIGN command.

The project is publicly available under the MIT License, and you are welcome to study and use the implementation and the underlying algorithm.

The original CATIA V5 project that served as the prototype is available here:

<https://github.com/ivashinpavel07/CATIA-V5-VBA-Align-Balloons>

CATIA demonstration:

<https://www.youtube.com/watch?v=UVtbpVKDkvY>

I would be very interested to know whether a geometry-aware or “Smart Align” mode for MLEADERALIGN could be considered for a future version of AutoCAD.

If this email has reached the Feedback Community administrators rather than the AutoCAD product team, I would appreciate it if you could forward it to the appropriate AutoCAD Product Manager or development team.

I would be happy to provide additional test drawings, algorithm details, source code information, or examples if needed.

Best regards,

Pavel Ivashin

ivashin.pasha@gmail.com

Nadia Rahman <nadia.rahman@autodesk.com>

17 сентября 2026 г. в 02:12

Кому: Pavel Ivashin <ivashin.pasha@gmail.com>

Копия: FeedbackCommunity <feedbackcommunity@autodesk.com>, AutoCAD Beta <autocad.beta@autodesk.com>

Hi Pavel,

Thank you for taking the time to share your proposal and supporting materials.

I shared your feedback with the AutoCAD team, and your idea has been entered for consideration as a potential future enhancement. We appreciate the detailed explanation and the effort you've invested in developing and documenting your solution.

While we can't comment on future product plans, feedback like yours is valuable and helps inform future discussions.

Thank you again for your contribution to AutoCAD.

Best regards,

Nadia Rahman

From: Pavel Ivashin <ivashin.pasha@gmail.com>

Sent: Saturday, September 12, 2026 2:10 PM

To: AutoCAD Beta <autocad.beta@autodesk.com>

Cc: FeedbackCommunity <feedbackcommunity@autodesk.com>

Subject: AutoCAD Feature Proposal — Smarter MLEADERALIGN Algorithm with Working AutoLISP Prototype

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Pavel Ivashin <ivashin.pasha@gmail.com>

17 сентября 2026 г. в 10:06

Кому: Nadia Rahman <nadia.rahman@autodesk.com>

Копия: FeedbackCommunity <feedbackcommunity@autodesk.com>, AutoCAD Beta <autocad.beta@autodesk.com>

Hi Nadia,

Thank you very much for your response and for sharing the proposal with the AutoCAD team.

I really appreciate that the idea has been entered for consideration as a potential future enhancement.

Since my original email, I have continued developing the prototype and have now released **MLeaderSmartAlign v0.7**.

The latest version is now fully standalone and combines both alignment modes into one interactive workflow:

Live Line → P2 → optional Live Perimeter

The user can first align MLeaders along a line with Live Preview, then either keep that result or continue directly into a four-side perimeter distribution mode.

Latest release:

<https://github.com/ivashinpavel07/AutoCAD-AutoLISP-MLeaderSmartAlign/releases/tag/v0.7>

Project repository:

<https://github.com/ivashinpavel07/AutoCAD-AutoLISP-MLeaderSmartAlign>

New video demonstration:

<https://youtu.be/nRvsLB-oLcl>

The video also shows part of the development process and then the resulting command being tested directly in AutoCAD.

I thought it would be useful to share the updated version in case the AutoCAD team reviews the proposal in more detail.

Thank you again for your time and for forwarding the idea internally.

Best regards,

Pavel Ivashin

ЧТ, 17 сент. 2026 г. в 02:12, Nadia Rahman <nadia.rahman@autodesk.com>:

[Цитируемый текст скрыт]

Nadia Rahman <nadia.rahman@autodesk.com>

18 сентября 2026 г. в 03:24

Кому: Pavel Ivashin <ivashin.pasha@gmail.com>

Копия: FeedbackCommunity <feedbackcommunity@autodesk.com>, AutoCAD Beta <autocad.beta@autodesk.com>

Hi Pavel,

Thank you for the update and for sharing the latest version of MLeaderSmartAlign.

I appreciate you sending over the updated materials, and I've shared them with the AutoCAD team for reference. Thank you again for your continued enthusiasm and for taking the time to share your work with us.

Kind regards,
Nadia Rahman

From: Pavel Ivashin <ivashin.pasha@gmail.com>

Sent: Thursday, September 17, 2026 12:06 AM

To: Nadia Rahman <nadia.rahman@autodesk.com>

Cc: FeedbackCommunity <feedbackcommunity@autodesk.com>; AutoCAD Beta
<autocad.beta@autodesk.com>

Subject: Re: AutoCAD Feature Proposal — Smarter MLEADERALIGN Algorithm with Working AutoLISP Prototype

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